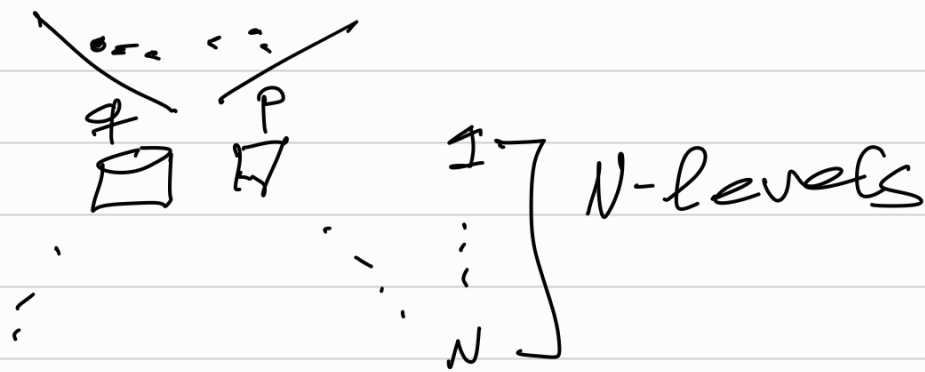


P : prob. of going Right
 q : prob. of going Left
 $P + q = 1$

• For a no-bias system: $p = q = 1/2$

• The position of a ball is given by X_N after N steps.

$$X = \sum_{j=1}^N \sigma_j \quad \sigma_j = \begin{cases} +1, & p \\ -1, & q \end{cases}$$

• Prob. of each level:

$$P_\sigma = p \delta_{\sigma,1} + q \delta_{\sigma,-1}$$

i.e. its a prob. density.

with $P_{\sigma_j} = P(\sigma_j)$

• Now, why are prob. multiplicative?

i.e.

$$P(\sigma_1, \dots, \sigma_N) = \prod_{i=1}^N P(\sigma_i)$$



i.e. if we follow $\rightarrow$ then at point B there's already an underlying prob. of $1/2$. This then has to be accounted for.

~~i.e.~~ $\rightarrow$ • has a prob. of 1. this is a "free" event. with no bias, the two options must sum to the 1 i.e.

• each red will have $1/2$ since there are two options. Again, continuing this to the next step

• Each has a prob of $1/4$

$\Rightarrow$ Hence, as we go along we "multiply" or, rather, we are splitting our odds

So, by the time we get to the end we have our value

$$P(\sigma_1, \dots, \sigma_N) = \prod_{j=1}^N P(\sigma_j)$$

* This is a 1D Random Walk.

* We can compute the average location a ball may be after N steps:

$$\langle X \rangle = \left\langle \sum_{j=1}^N \sigma_j \right\rangle$$

Recall, the average is given by:

$$\langle x \rangle = \frac{P(x_1)x_1 + P(x_2)x_2 + \dots + P(x_N)x_N}{P(x_1) + \dots + P(x_N)}$$

$$\langle x \rangle = \frac{\sum_{i=1}^N P(x_i) x_i}{\sum_{i=1}^N P(x_i)}$$

x : can take values $x_1, \dots, x_N$
 Now, the average of a function

$$\langle f(x) \rangle = \frac{\sum_{i=1}^N P(x_i) f(x_i)}{\sum_{i=1}^N P(x_i)}$$

Backtrack a bit to the notation of the random walk:

$$X = mL \quad -L \xrightarrow{\quad L \quad} R \quad / \quad -N \leq x \leq N$$

$P_N(m)$:

$$n_1: \rightarrow \quad / \quad \text{Total steps: } N = n_1 + n_2$$

$$n_2: \leftarrow$$

Net displacement $m \equiv n_1 - n_2$

$$m = n_1 - n_2 = n_1 - (N - n_1)$$

$$m = 2n_1 - N$$

Claim: if N is odd, then m must also be odd

proof: If N is odd then $\exists \alpha \in \mathbb{N}$ s.t. $N = 2\alpha + 1$

$\therefore$

$$m = 2n_1 - 2\alpha - 1 = 2(n_1 - \alpha) - 1$$

Since $n_1 \in \mathbb{Z}$ and $\alpha \in \mathbb{Z}$, then by closure of $\mathbb{Z}$

$$n_1 - \alpha \in \mathbb{Z} = \beta$$

likewise,

$$n_2 = 2\alpha - 1 + 1 - 1 = 2\alpha - 1$$

$$m = 2(\beta - 1) + 1$$

$$\beta - 1 \in \mathbb{Z} \equiv \beta'$$

$$m = 2\beta' + 1$$

likewise, if N is even, so too is m

* The total prob., as above, is the product of each step

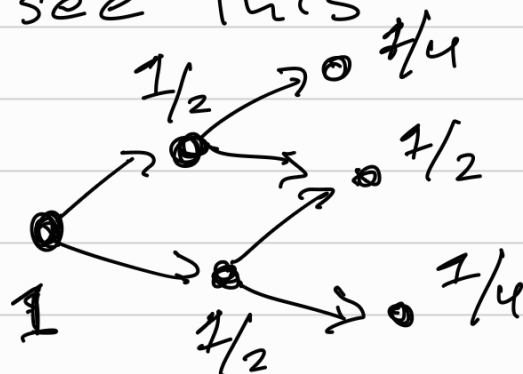
$$\underbrace{P \cdots P}_{n_1} \underbrace{q \cdots q}_{n_2} = P^{n_1} q^{n_2}$$

Now, we know there are multiple ways of ending up at the same point. i.e. multiple paths have the same probability.

$$N = n_1 + n_2$$

Given by $\binom{N}{n_1} = \frac{N!}{n_1! n_2!}$

* So, what does this mean, well, see this



These are not $1/3$.

the original prob. is $1/4$ but there are 2 paths

$$1 + 1 = 2$$

$$P_2(1/2) = 2 \cdot (1/2) = 1$$

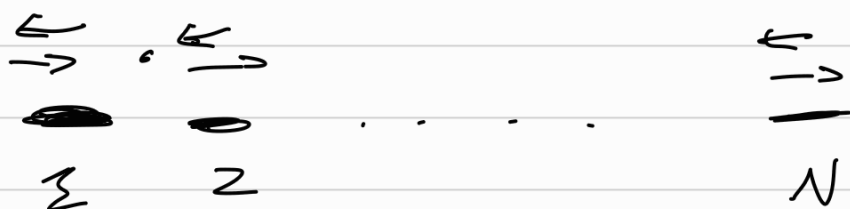
* Thus, the probability of getting to a point N , with no bias, is

$$P_N(n_1) = \frac{N!}{n_1! n_2!} p^{n_1} q^{n_2} = \binom{N}{n_1} p^{n_1} q^{n_2}$$

Reasoning behind $N! / n_1! n_2!$:

1st choice: 1 of the N
 2nd " " " 1 of the $(N-1)$
 ⋮
 ⋮

N^{th} " " 1 of 1



All N can be occupied in $N!$ different ways

$$N! = N(N-1) \dots 1$$

$\Rightarrow$ This is for Distinguishable objects but at each level the moments are not related/distinguishable. Each ending position, you cannot tell the exact path.

$\Rightarrow$ all n_1 permutations lead to the same outcome and same for n_2 .
To get rid of these redundancies

$$N! \rightarrow \frac{N!}{n_1! n_2!}$$

Binomial expansion:

$$(p+q)^N = \sum_{n=1}^N \binom{N}{n} p^n q^{N-n}$$

$$* P_N(m) = W_N(n_1)$$

$$\begin{array}{l|l} n_1 + n_2 = N & n_1 = m + n_2 \\ n_1 - n_2 = m & = m + N - n_1 \end{array}$$

$\therefore$

$$n_1 = \frac{1}{2}(m+N)$$

$$n_2 = \frac{1}{2}(N-m)$$

$$\text{Now, } W_N(n_1) = \frac{N!}{n_1! n_2!} p^{n_1} q^{n_2}$$

$$P_N(m) = \frac{N!}{\frac{1}{2}(m+N)! \frac{1}{2}(N-m)!} p^{\frac{1}{2}(m+N)} (1-p)^{\frac{1}{2}(N-m)}$$

• In the case of $p=q=1/2$

$$P_N(m) = \frac{N!}{[(N-m)/2]! [(N+m)/2]!} \left(\frac{1}{2}\right)^N$$

ex: $p=1/2=q$ and $N=3$

$n_1 = 0, 1, 2, \text{ or } 3$

$$\underline{A_1 = 0} : 0 = \frac{1}{2}(3+m) \rightarrow m = -3$$

$$P_3(-3) = \left(\frac{1}{2}\right)^3 = 1/8$$

$$\underline{n_1 = 1} \quad 2 = 3 + m \Rightarrow m = -1$$

$$P_3(-1) = \frac{3!}{1!2!} \left(\frac{1}{2}\right)^3 = \underline{\underline{\frac{3}{8}}}$$

$$4 = 3 + m \Rightarrow m = 1$$

$$P_3(1) = \frac{3!}{2!1!} \left(\frac{1}{2}\right)^3 = \underline{\underline{\frac{3}{8}}}$$

$$\underline{n_3 = 3} \quad \frac{3!}{0!3!} \left(\frac{1}{2}\right)^3 = \underline{\underline{\frac{1}{8}}}$$

Mean Values: